CS5002
SP 2026
Basics of Propositional Logic

Problem 1. (**15 points**) Translate the following symbolic propositions into English sentences. If you would like to simplify the propositions before creating a sentence, include the simplified symbolic logic along with your sentence.

Ex. $\neg P \wedge \neg Q$ might translate to *it is not raining and it is not snowing*, or you could simplify the logic to $\neg(P \vee Q)$ and then translate to *it is not raining or snowing*

- $\neg\neg\neg P$
- $P \wedge Q \wedge R$
- $(Q \vee R) \wedge \neg(Q \wedge R)$
- $\neg((P \vee Q) \wedge (\neg Q \vee R))$
- $(P \bigwedge Q) \bigwedge (\neg Q \bigwedge R)$

Problem 2. (**10 points**) A *tautology* is a statement that is always true, regardless of what any values in the proposition are. Using a truth table, prove whether or not $(P \wedge Q) \vee (P \vee Q)$ is a tautology.

Problem 3. (**25 points**) On the land of Knights and Knaves, everyone is either a Knight or a Knave. Knights can only speak in true statements while Knaves can only lie.

You meet three people named Weijian, Xuetao, and Yi.

- Weijian says: "Xuetao is a Knave".
- Xuetao says: "Yi is a Knave".
- Yi says: "Weijian and Xuetao are both Knaves".

Determine the status of all 3 people (knight or knave). Prove that you are correct using a truth table using the following propositions:

- W = Weijian is a Knight
- X = Xuetao is a Knight
- Y = Yi is a Knight

Problem 4. (**20 points**)
For this problem, we'll be making a truth table for a vending machine. A user can purchase a bottled water by inserting one dollar into the machine. If the machine has no more water bottles, the machine will return the dollar back to the user. Otherwise, as soon as the machine has received the money, a button will flash. The user needs to press that button to receive a bottled water. If the button is not pressed, the machine returns the dollar back to the user.

There are three possible options:

- **N** - The machine has **no more** water bottles.
- **P** - The correct amount of money has been input into the machine.
- **B** - The button has been pushed.

The vending machine may take two actions:

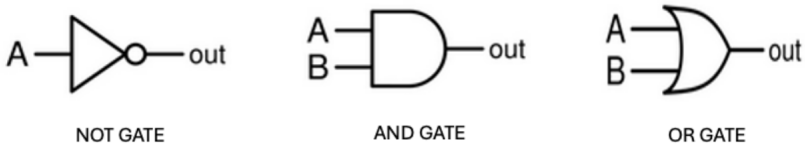
- **G** - Give 1 water bottle.
- **R** - Return money.

N	**P**	**B**	**G**	**R**
T	T	T	?	?
T	T	F	?	?
T	F	T	?	?
T	F	F	?	?
F	T	T	?	?
F	T	F	?	?
F	F	T	?	?
F	F	F	?	?

Identify which two variables (of N, P, B, G, and R) are logically equivalent.

Problem 5. (**30 points total**) **An Introduction to NAND:**

In class, we talked about how AND, OR, and NOT are all simple formulations of circuits:



This homework will be about a new type of gate formulation: NAND This is a shorthand of NOT AND (or $\neg(A \wedge B)$)

A	B	$A \wedge B$	A NAND B
T	T	T	F
T	F	F	T
F	T	F	T
F	F	F	T

A circuit diagram for NAND looks like this:

It should look pretty similar to an AND but there is an additional small circle before the right hand line.

As we talked about in class, every truth table, Boolean formula, and circuit can be implemented using just AND, OR, and NOT gates operators; hence, the collection AND, OR, NOT is what we call "logically complete".

In this problem, you will show that the NAND gate, by itself, is logically complete too.

To do so, we need to prove that we can make AND, OR, and NOT gates using just NAND gates

(a) **Gate 1 (3 points)** - Fill in the following truth table:

P	P NAND P
T	
F	

(b) (**2 points**) Which of the 3 operators (AND, OR, and NOT) is logically equivalent to P NAND P?

(c) **Gate 2 (10 points)** - The next statement we'll be examining is $\neg P$ NAND $\neg Q$:

But thinking through that statement will be a lot easier if we first transform it into something more tractable. or an equivalence proof.

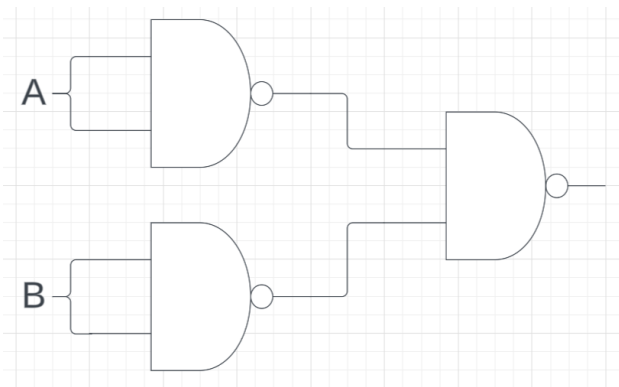
Using an equivalence proof, turn $\neg P$ NAND $\neg Q$ into a statement that only uses AND and OR operators (no NOTs).

Reminder an equivalence proof is when we use the laws of equivalence to transform 1 statement into a logically equivalent statement.

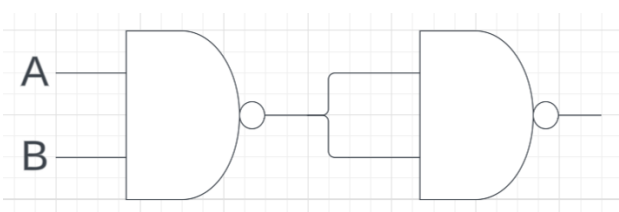
Try to get your final answer as simple as possible. Indicate all logical equivalence laws used to get from $\neg P$ NAND $\neg Q$ to your answer.

(d) (**5 points**) Which of the 3 operators (AND, OR, and NOT) is logically equivalent to $\neg P$ NAND $\neg Q$?

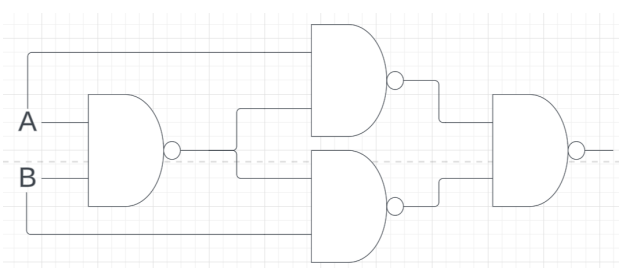
(e) **Gate 3 (10 points)** - We've so far replicated the behavior of 2 of our big 3 operators from class. By process of elimination, we only have 1 gate left to derive so you probably know what this gate is.
What is that final operator? Which of the following circuit diagrams replicates the behavior of our final operator (Gate 1, 2, or 3)? Prove you're correct either with a truth table **or** through equivalence proof.



Gate 1



Gate 2



Gate 3